1 Simulation component: social information and variability of foraging gain

Hypothesis. Social information reduces variability in foraging gain, mainly when krill is patchy and ephemeral and abundant enough to be shared.

Prediction. Within-whale relative variability of intake (coefficient of variation, CV) and the probability of a shortfall fall as information availability rises. Raw variance is not the right metric (see below), and the effect should weaken or reverse when krill is scarce.

1.1 Setup

Consider a foraging bout of length B (a day or dive bout). A whale alternates between searching and feeding. While searching it finds a patch at rate λ. While feeding it gains intake at rate r, and feeding ends when the patch disappears, which happens at rate $\delta = 1/L$, where L is the mean patch lifetime. Large L means persistent patches and small L means ephemeral patches.

Personal information gives a baseline finding rate λ_0. Social information adds a second route to a patch. Call availability $q \in [0, 1]$ is the probability that a searching whale receives a call, and the information is discounted because the patch may have vanished by the time the whale arrives. If travel to the caller takes time τ, the probability that the patch is still there is $e^{-\tau/L}$, so

$$\lambda(q) = \lambda_0 + \kappa\, q\, e^{-\tau/L}, \tag{1}$$

where κ is the rate at which calls would lead to a patch if patches never vanished. The factor $e^{-\tau/L}$ is the scale-matching problem in compact form: information transfer is only useful when its travel time is short relative to the lifetime of the resource.

1.2 Agent-based simulation

An agent-based model checks whether this trend emerges when searching, patch turnover and competition are simulated explicitly (code: `abm_social_info_robust.py`).

World. A 100×100 torus with $K = 8$ patches (radius 3). Each patch disappears with probability $1/L$ per step and reappears at a random location. **Agents.** $N = 20$ foragers move at one cell per step. A searching agent that comes within 5 cells of a patch heads to it; otherwise it does a correlated random walk. Feeding agents stay on their patch and act as callers. **Social information.** A searching agent with no current target hears the nearest caller within 40 cells with probability q per step, stores that location, and walks to it, even if the patch has vanished on arrival. **Competition.** Agents on one patch share intake as $1/n^c$, with $c = 0$ (abundant, shareable) or $c = 1$ (scarce). **Outcome.** Intake per bout of 200 steps, summarised as mean, within-agent variance, CV, and the proportion of bouts below a threshold of 50 intake units. The first 500 steps are burn-in, and accumulated intake is reset when measurement begins so that no recorded bout includes burn-in. Values are means of 6 replicate runs of 4500 steps, and the code validates its parameters before running.

Results.

- *Abundant krill ($c = 0$).* Increasing call availability reduced CV across all patch lifetimes (e.g. 0.77 to 0.30 at $L = 80$) and reduced shortfall probability (0.38 to 0.03). Mean intake also increased. Raw variance did not always decrease, highlighting why CV and shortfall probability are more appropriate measures of relative variability.

- *Scarce krill ($c = 1$).* Social information concentrated agents on a limited number of patches. Mean intake declined, shortfall probability increased, and CV was flat or increased (e.g. 0.80 to 0.98 at $L = 300$).

	Abundant ($c = 0$)		Scarce ($c = 1$)	
Mean patch life	CV, $q = 0 \to 1$	Shortfall	CV, $q = 0 \to 1$	Shortfall
20	$0.90 \to 0.54$	$0.75 \to 0.41$	$0.90 \to 0.90$	$0.87 \to 1.00$
80	$0.77 \to 0.30$	$0.38 \to 0.03$	$0.88 \to 0.91$	$0.68 \to 0.98$
300	$0.49 \to 0.17$	$0.15 \to 0.01$	$0.80 \to 0.98$	$0.60 \to 0.97$
1500	$0.23 \to 0.05$	$0.04 \to 0.00$	$0.66 \to 0.69$	$0.57 \to 0.94$

Table 1: Simulated CV of bout intake and probability of a shortfall bout without ($q = 0$) and with ($q = 1$) social information, by patch lifetime and abundance. Means of 6 runs.

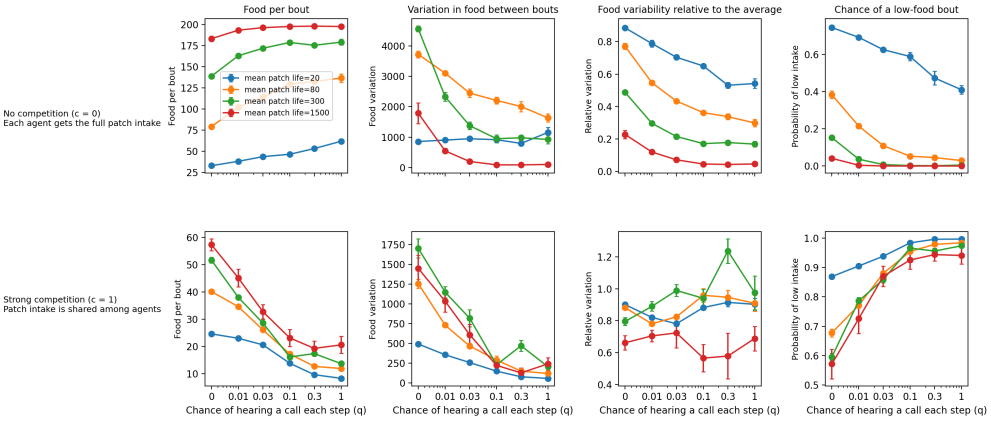


Figure 1: Simulation results showing how social information affects foraging gain and variability.

Interpretation. Social information reduced relative variability in the abundant, shareable regime, but its benefits disappeared or reversed under strong competition. These results qualitatively support the prediction that social information can improve foraging outcomes when resources are patchy and shareable. Parameter values were illustrative rather than empirically calibrated.

Mapping to whale data. Call availability (q) corresponds to received call rate, patch lifetime (L) to krill swarm persistence, and competition (c) to resource shareability. The empirical comparison would examine within-whale CV and the proportion of low-lunge bouts against call exposure, using lunges per bout as a proxy for foraging gain.

Supplementary materials. The Python implementation of the agent-based model is available here. The corresponding simulation animation is available here.